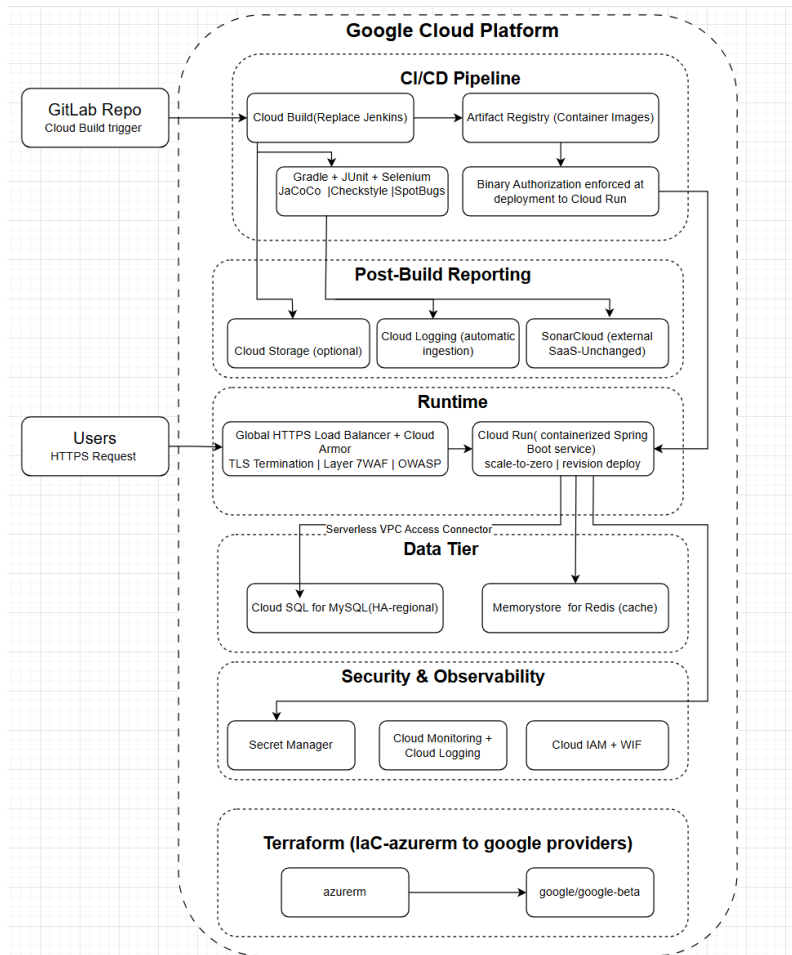


Research Report(750 Words)

1. Assumptions

- Moderate initial load (~1000–10,000 monthly active users) with growth potential
- CI/CD pipeline remains GitLab-triggered

2. Proposed Architecture



Google Cloud Platform (GCP) architecture replaces two-VM model with a managed, container-native approach to reduce operational overhead while improving scalability, reliability, and separation between CI and runtime environments.

Gitlab triggers *Cloud Build*[2], executing the existing Gradle pipeline in containerised steps. Built images are stored in *Artifact Registry* which integrates natively with *Cloud Run*[1]. Supply chain security is enforced using *Binary Authorization*: Cloud Build produces cryptographic attestations that are verified at deployment, ensuring only trusted images are released[3][1].

Application is deployed on *Cloud Run*, a serverless platform suited for web and event-driven microservice architectures[1][2]. It automatically scales with demand and scales to zero when idle, eliminating unnecessary compute cost and improving efficiency[4][1]. Traffic is routed through *Cloud Load Balancing* with TLS and Cloud Armor. *Cloud Armor* provides Layer 7 filtering and scrubs incoming requests for common web attacks using OWASP-based WAF at the network edge before traffic reaches the VPC[5].

Persistent storage is handled by *Cloud SQL(MySQL)*, providing high availability[2].

Credentials are managed via Secret Manager and Workload Identity Federation, replacing the SSH key bridge between VM1 and VM2[3]. Cloud Monitoring and Cloud Logging provide unified observability across all pipeline and runtime components.

3. Pipeline Migration

DTL Component	GCP Equivalent
GitLab → Jenkins trigger	GitLab → Cloud Build trigger
Gradle build + JUnit/Selenium tests	Cloud Build(same gradlew) , Selenium(Chrome-enabled build image)
Code quality tools	Cloud Build(unchanged)
Docker build + Hadolint + Trivy	Cloud Build + Binary Authorization[3]
SCP to VM2	Artifact Registry → Cloud Run
Docker run on VM2	Cloud Run revisions
Nginx reverse proxy	Cloud Load Balancer + Cloud Armor[5]
SSH keys/Jenkins creds	Workload Identity + Secret Manager[3]
Terraform (azurerms)	Terraform (google) ~70% reuse[10]

4. Critical Evaluation

4.1. Compute: Cloud Run vs GKE vs Compute Engine

Three deployment options were evaluated: **Compute Engine (VMs)** enables reuse of existing Terraform scripts but retains operational overhead (patching, scaling, security management), resulting in the highest total cost of ownership[6]. **GKE (Google Kubernetes Engine)** provides full orchestration and container control[2] but requires solid Kubernetes administration, security, and development skills[6], increasing cost and introducing unnecessary complexity for the expected 1,000–10,000 MAU. **Cloud Run** is selected: stateless Spring Boot application backed by Cloud SQL and Memorystore enables auto-scaling, removes infrastructure overhead, and supports secure CI/CD natively[1][2]. **GKE** remains a future option if complexity or traffic increases[6].

4.2. Cloud Build vs Jenkins

While retaining Jenkins allows reuse of existing pipeline scripts and tools (Gradle, SonarCloud) with minimal changes, it requires managing a server on GCP. Migrating to Cloud Build removes VM maintenance overhead while supporting all pipeline stages (build, test, scan, deploy), with native integration to Artifact Registry and Cloud Run[2].

4.3. Cost Considerations

service_display_name	name	quantity	region	service_id	sku	total_price, USD
Artifact Registry	Artifact Registry Storage	1	global	149C-F9EC-3994	8502-299A-ABAF	0.04313
Cloud Run (Cloud Run)	Services CPU Tier 2 (Request-based billing)	100000000	global	152E-C115-5142	065C-4237-027A	3.36
Cloud Run (Cloud Run)	Services Memory Tier 2 (Request-based billing)	50000000	global	152E-C115-5142	600C-3782-6706	0.175
Cloud Run (Cloud Run)	Requests	250000	global	152E-C115-5142	2DA5-55D3-E679	0
MySQL (Cloud SQL)	Cloud SQL for MySQL: Zonal - Small instance in London	730	eu-west-2	9662-B51E-5089	A6A8-E03A-4359	30.66
MySQL (Cloud SQL)	Cloud SQL for MySQL: Regional - Standard storage in London	7300	eu-west-2	9662-B51E-5089	B3E0-708F-34BF	3.7998
Memorystore for Redis (Cloud Memorystore)	Redis Capacity Basic M1 London	730	eu-west-2	54F5-2C11-0467	8FF9-D19D-3185	36.02867
Cloud Load Balancing (Networking)	Global External Application Load Balancer Inbound Data Processing for London (eu-west-2)	1		E505-1604-58F8	C44D-AEFB-B855	0.00931
Cloud Load Balancing (Networking)	Global External Application Load Balancer Outbound Data Processing for London (eu-west-2)	1		E505-1604-58F8	E8B5-84A5-943D	0.00931
Cloud Load Balancing (Networking)	Regional Internal Proxy Network Load Balancer Inbound Data Processing for London (eu-west-2)	1		E505-1604-58F8	0045-6DE4-DA42	0.00931
Cloud Load Balancing (Networking)	Cloud Load Balancer Forwarding Rule Minimum Global	1		E505-1604-58F8	DEE3-C42E-3E4D	18.25
Cloud Armor	Networking Cloud Armor Requests	250000	global	E505-1604-58F8	1A07-0EB9-C4BE	0.1875
Cloud Armor	Networking Cloud Armor Policy	1	global	E505-1604-58F8	4B13-E64F-4A2B	5
Cloud Armor	Networking Cloud Armor Rule	5	global	E505-1604-58F8	A321-89BD-F5BC	5
					Total Price:	102.53624

Exchange rate: 1 USD = £0.73[8]

Full cost breakdown[7] shows costs are driven by fixed services, Cloud SQL(~£25) and Memorystore(~£26), which run continuously, while Cloud Run costs ~£2.58/month due to scale-to-zero. This creates a predictable model where costs scale with usage with stable fixed baselines. Limitation is vertical scaling in Cloud SQL, requiring brief downtime; at higher scale, migration to Cloud Spanner or Google Kubernetes Engine with committed-use discounts may be more cost-efficient.

4.4. Security

Proposed architecture improves on the DTL model which relied on static SSH keys and Jenkins credentials[2]. The main residual risk is IAM misconfiguration, which could expose services like Cloud SQL or Memorystore; this is mitigated through least-privilege roles, audit logging, and staged testing. A limitation is that Cloud Armor Standard lacks advanced threat protection, which requires the Enterprise tier at additional cost[11].

4.5. Lock-in and Portability

Cloud Build and IAM introduce moderate GCP lock-in, but portability is preserved through OCI-compliant Docker images[9], MySQL-compatible Cloud SQL, and provider-agnostic Terraform[10], enabling migration to other cloud platforms if needed.

4.6. Agile Development

Cloud Run's revision-based deployments enable PR preview environments and canary releases via Cloud Build, routing small traffic percentages to new versions with instant rollback if needed[4][1], improving feedback loops and supporting continuous delivery.

5. Migration Strategy

Parallel approach: deploy GCP via Terraform alongside DTL; migrate the database with read-only sync window; run Cloud Build and Jenkins in parallel; deploy to Cloud Run and validate; shift traffic gradually(10%→100%); decommission DTL after ~48-hour monitoring. Risks include data inconsistency, cold starts, and pipeline failure, mitigated by sync controls, minimum instances, and Jenkins fallback.

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